THE CHALLENGE · UNDERSTAND AN UNKNOWN MACHINE

The machine was fluent. We needed the dictionary.

CODES, BUT WHICH ACTIONS?

```
03  →  ?
04  →  ?
19  →  ?
```



A “stack machine”

It stacks values,
then copies them, moves them
or performs calculations.

The helpful clue: 4 sample programs + their expected outputs

THE TAKEAWAY

The examples will help us **recover the meaning of the codes.**

UNDERSTAND · A SMALL EXAMPLE SHOWS THE PRINCIPLE

Copy or swap?
The output tells them apart.

INVENTED TEACHING EXAMPLE · THE REAL TESTS COME NEXT

STARTING STACK

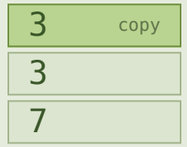


We process the top value first.

Unknown instruction
then print the top

HYPOTHESIS A

Copy the top

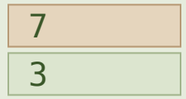


→ OUTPUT 3

✓ consistent

HYPOTHESIS B

Swap the two



→ OUTPUT 7

× rejected

IF THE EXPECTED OUTPUT IS 3 “Swap” cannot be the right instruction.

THE TAKEAWAY

A known example helps us **reject an interpretation that does not fit.**

THE SOLVE · APPLY THIS TEST TO EVERY DICTIONARY

The examples narrow it down. One dictionary remains.

01

First, the stack operations

Try mappings between codes and actions.

Example 1 Example 2

1 mapping

02

Then, the arithmetic operations

Keep mappings that produce the right outputs.

Example 3 Example 4

1 mapping

346 possibilities after example 3 → **1** after example 4

Using the correct interpretation of signed numbers.

THE TAKEAWAY

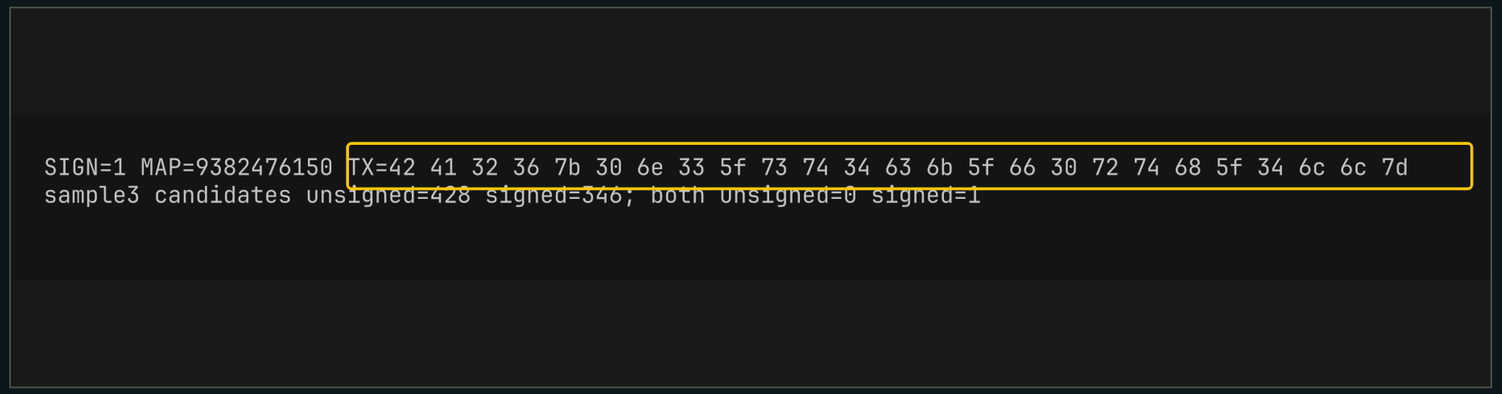
The unknown transmission is read **using this verified dictionary.**

THE EVIDENCE · THE RECOVERED DICTIONARY WORKS

The machine gets its language back. The message becomes readable.

03	copy the top
04	emit a byte
19	add

Three entries from the recovered dictionary.
Codes are written in hexadecimal.



SAVED EVIDENCE · CLICK TO ENLARGE

Solved by: **Codex** · Saved solver output.

`BA26{0n3_st4ck_f0rth_4ll}`

THE TAKEAWAY

The known outputs became **a guide to reading the unknown.**